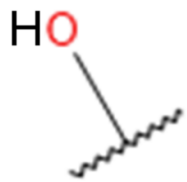
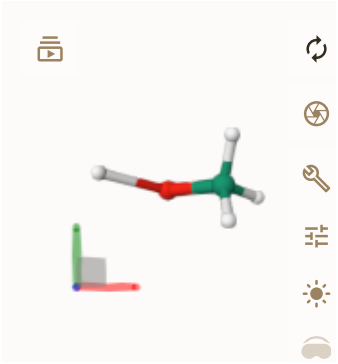
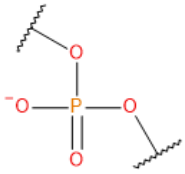
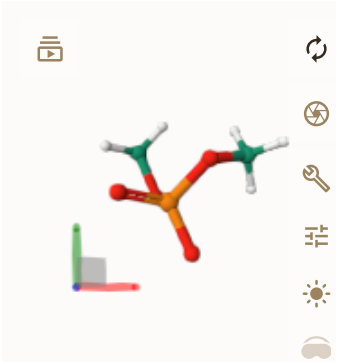
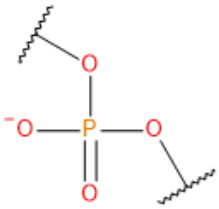
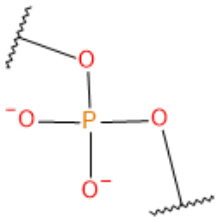
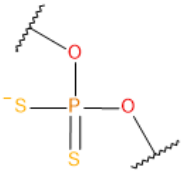
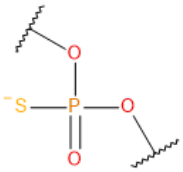
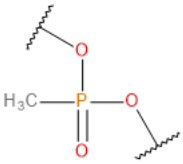
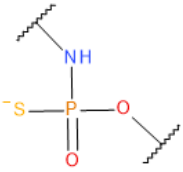
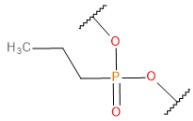
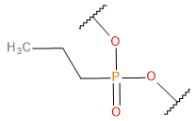
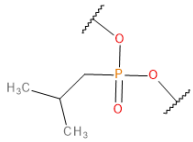
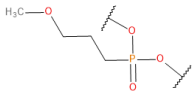
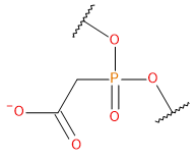
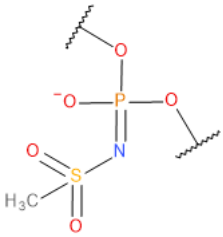
Catalog

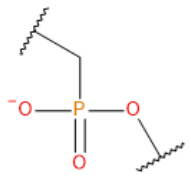
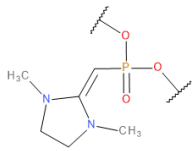
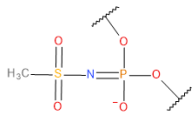
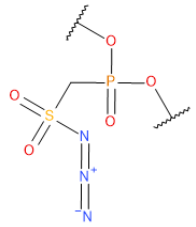
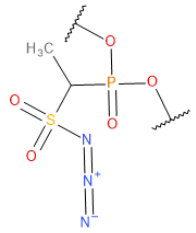
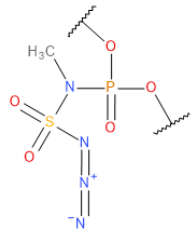
Backbone fragment library

Assume that when it is a stereo-isomer, the fragment will have the final number 1 for the (S) configuration and 2 for the (R) configuration.

FRAGMENT ID	2D Structure	Name	Notes	3D mol*
5PO		5PO	5′-terminal cap	
DPO		DPO	Standard dimethyl phosphate (regular backbone) Charge -1 Atom types from OL15/21	

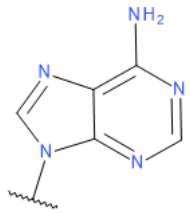
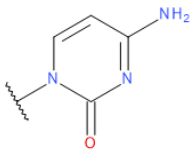
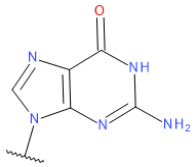
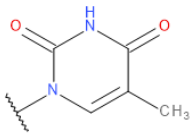
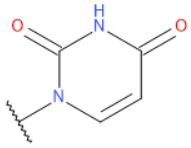
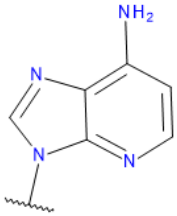
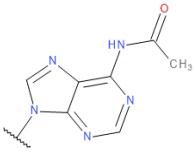
RPO		RPO	Standard dimethyl phosphate (regular backbone) Charge -1 Atom types from OL3	
BPO		BPO		
PSS		PSS	Phosphorodithioate Charge -1	
PS1 PS2		PS1 (S) PS2 (R)	Thiophosphate Charge -1 Other names: phosphorothioate	
MP1 MP2		MP1 (S) MP2 (R)	Methylphosphate	
NS1 NS2		NS1 (S) NS2 (R)	Thio phosphoramidate Charge -1	

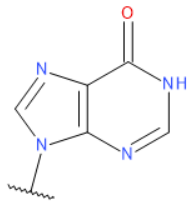
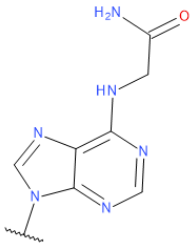
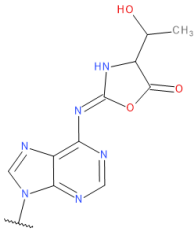
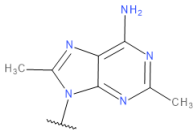
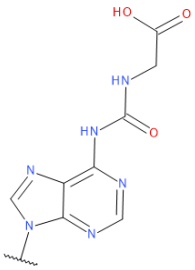
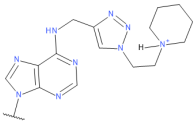
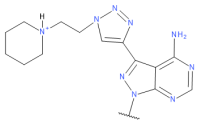
PR1		PR1	Propyl backbone (S)	
PR2		PR2	Propyl backbone (R)	
IB1 IB2		IB1 (S) IB2 (R)	Isobutyl backbone	
MO1 MO2		MO1 (S) MO2 (R)	MOP backbone	
PC1 PC2		PC1 (S) PC2 (R)	Phosphonoacetate Charge -1	
MES		MES		

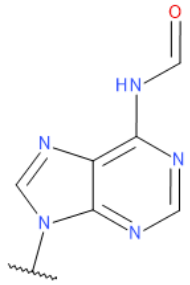
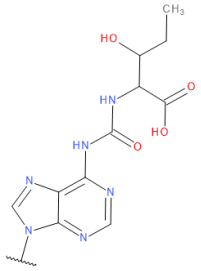
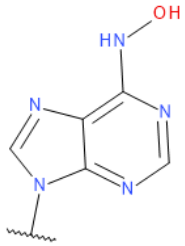
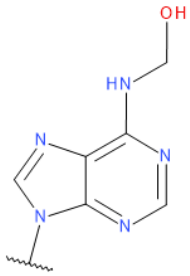
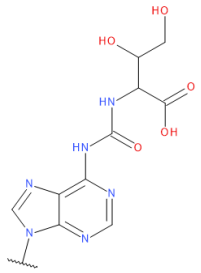
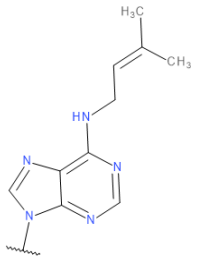
MXG		MXG	Methylene phosphonate Charge -1	
PN1 PN2		PN1 (S) PN2 (R)	Phosphoryl guanidine	
WX1 WX2		MX1 (S) MX2 (R)	Mesyl -N-phosphonate Charge -1	
VX1 VX2		VX1 (S) VX2 (R)	Ethyl-mesyl phosphonate	
XV1 XV2		XV1 (S) XV2 (R)	Isopropyl mesyl phosphonate	
ZV1 ZV2		ZV1 (S) ZV2 (R)	Amine-mesyl phosphonate	

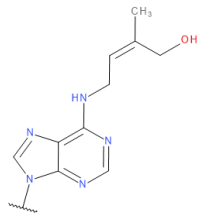
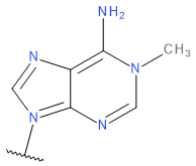
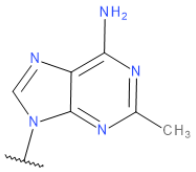
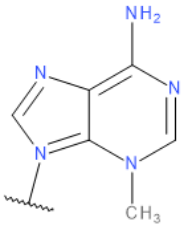
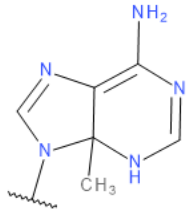
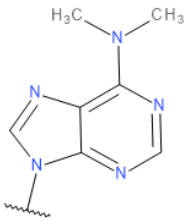
Base fragment library

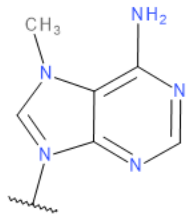
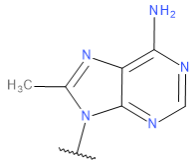
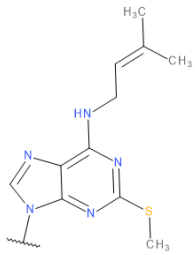
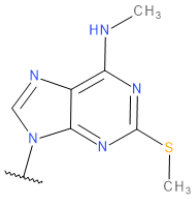
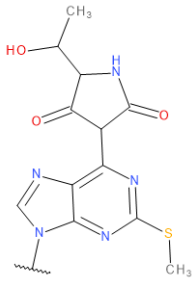
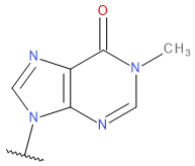
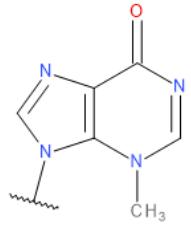
FRAGMENT ID	New 2D	Name	Notes
Canonical Nucleobases			

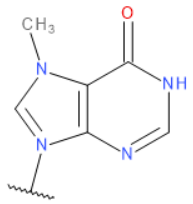
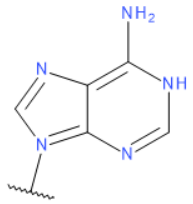
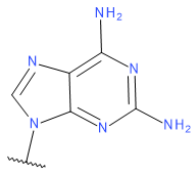
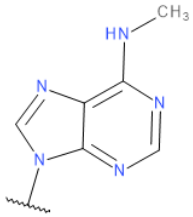
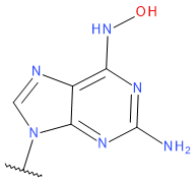
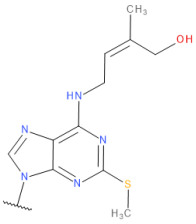
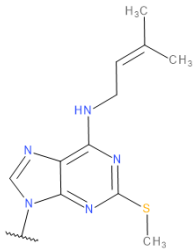
DAA/RAA		Adenine	Ideal module Name HARDCODED Charges from OL3/OL15
DCC/RCC		Cytidine	Ideal module Name HARDCODED Charges from OL3/OL15
DGG/RGG		Guanine	Ideal module Name HARDCODED Charges from OL3/OL15
DTT		Thymine	Ideal module Name HARDCODED Charges from OL3/OL15
RUU		Uracil	Ideal module Name HARDCODED Charges from OL3/OL15
Modified Adenine			
1DP		N1-deaza-adenosine-5′-monophosphate	
A6A		N6-acetyladenosine	

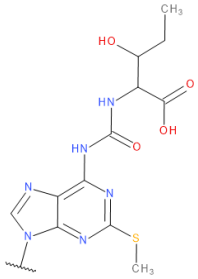
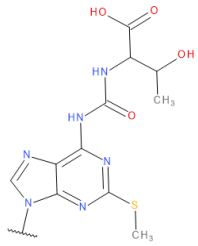
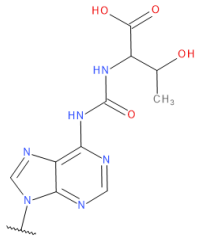
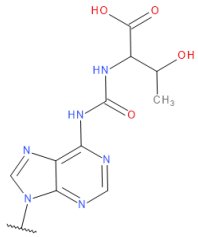
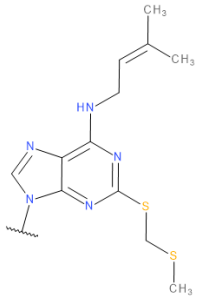
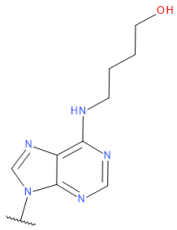
BII		Inosine	
C6A		N6-carbamoylmethyl adenine	
CTA		cyclic N6-threonylcarbamoyladenosine	
DMA		2,8-dimethyladenosine	
G6A		N6-glycinylcarbamoyladenosine	
E6A		N-ethylpiperidine-6’triazole modified adenine	+1 charge
E7A		N-ethylpiperidine-7’-EAA triazole modified adenine	+1 charge

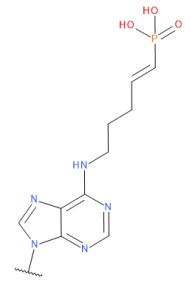
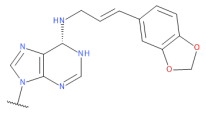
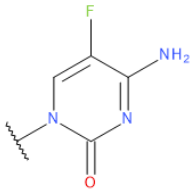
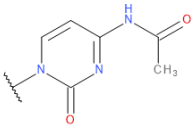
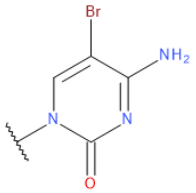
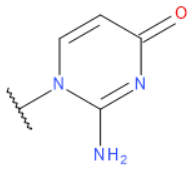
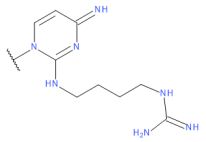
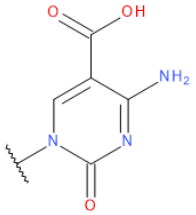
F6A		N6-formyladenosine	
H6A		N6-hydroxynorvalylcarbamoyladenosine	
HNA		6-hydroxyaminopurine	
HMA		N6-hydroxymethyladenosine	
HTA		Hydroxy-N6-threonylcarbamoyladenosine	
I6A		N6-isopentenyladenosine	

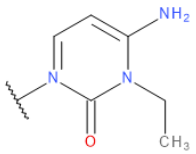
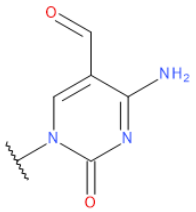
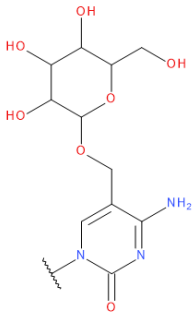
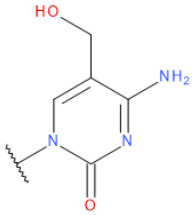
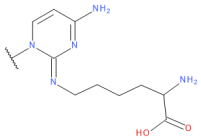
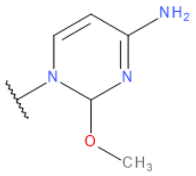
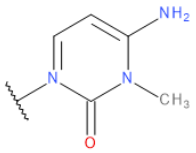
IGA		Isoguanosine	
IOA		N6-(cis-hydroxyisopentenyl)adenosine	
M1A		1-methyladenosine	
M2A		2-methyladenosine	
M3A		3-methyladenosine	
M4A		4-methyladenosine	
M6A		N6,N6-dimethyladenosine	

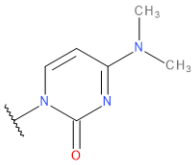
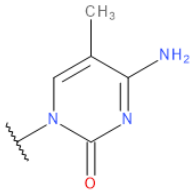
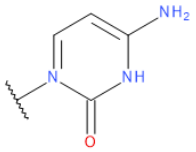
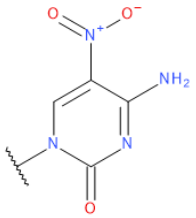
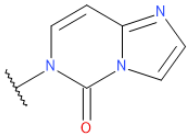
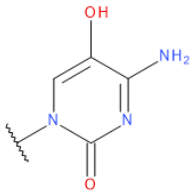
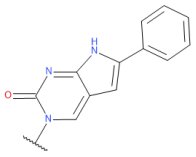
M7A		7-methyladenosine	
M8A		8-methyladenosine	
MIA		2-methylthio-N6-isopentenyl-adenosine-5′-monophosphate	
MSA		2-methylthio-N6-methyladenosine	
MTA		2-methylthio cyclic N6-threonylcarbamoyladenosine	
M1I		1-methylinosine	
M3I		3-methylhypoxyanthine	

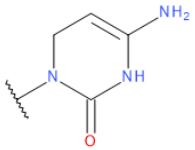
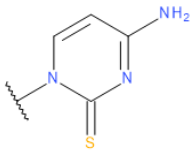
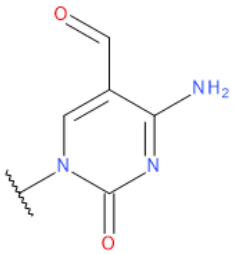
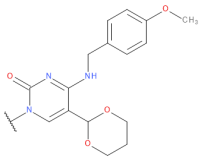
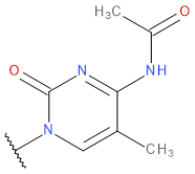
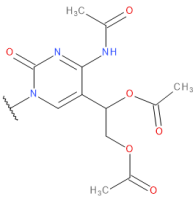
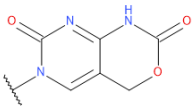
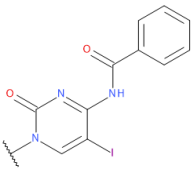
M7I		7-methylhypoxanthine	
N1A		N1-protonated adenosine	+1 charge
N2A		2-aminoadenine	
N6A		N6-methyladenosine	
NHA		2-amino-6-hydroxyaminopurine	
S6A		2-methylthio-N6-(cis-hydroxyisopentenyl) adenosine	
SIA		2-methylthio-N6-isopentenyladenosine	

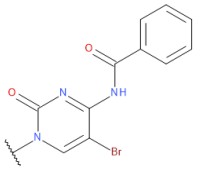
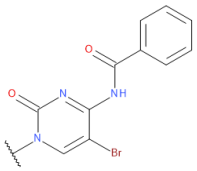
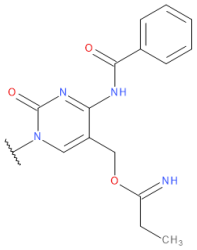
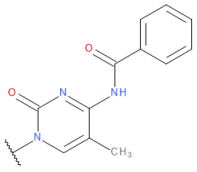
SHA		2-methylthio-N6-hydroxynorvalylcarbamoyladenosine	
STA		2-methylthio-N6-threonylcarbamoyladenosine	
T6A		N6-methyl-N6-threonylcarbamoyladenosine	
TCA		N6-threonylcarbamoyladenosine-5′-monophosphate	
TMA		2-methylthiomethylenethio-N6-isopentenyl-adenosine	
TMB			

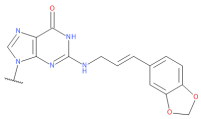
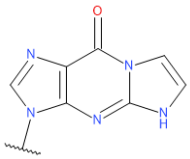
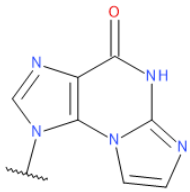
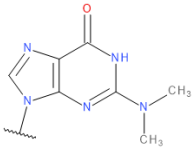
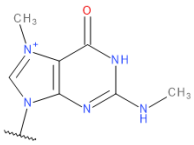
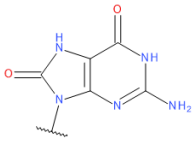
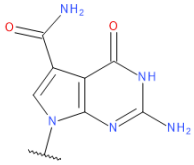
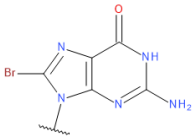
TMC			
TIA		N2-trans-isosafrole-dA-adduct	
Modified Cytidine			
5FC		5-fluorocytidine 5′-(dihydrogen phosphate)	
A4C		N4-acetylcytidine	
B5C		5-bromocytidine 5′-(dihydrogen phosphate)	
BIC		isocytidine-5′-monophosphate	
BPC		agmatidine	
C5C		5-carboxylcytosine	

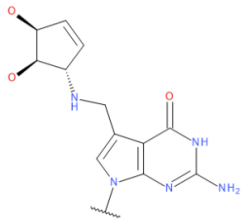
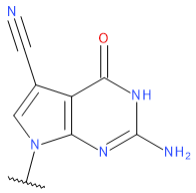
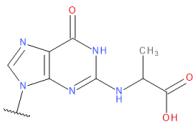
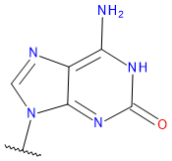
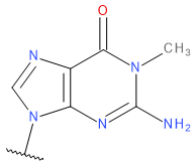
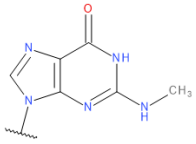
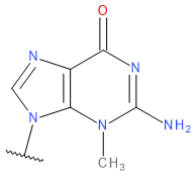
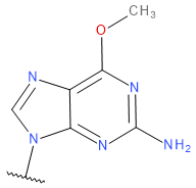
E3C		3-ethylcytosine	
F5C		5-formylcytosine	
G5C		5-glucosylmethylcytosine	
H5C		5-hydroxymethylcytosine	
K2C		2-lysidine	
M2C		2-O-methylcytosine	
M3C		3-methylcytosine	

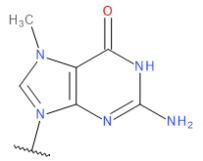
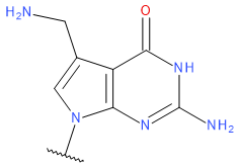
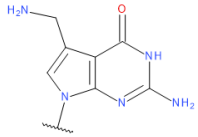
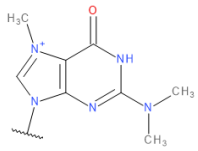
M4C		N4,N4-dimethylcytidine	
M5C		5’-methylcytidine	
N3C		N3-protonated cytidine	Charge +1
N4C		N4-methylcytosine	
N5C		5-nitrocytidine 5′-(dihydrogen phosphate)	Charge +1
NEC		3,N4-ethenocytosine	
O5C		5-hydroxycytidine	
PPC		6’-phenylpyrrolo-cytosine	

RYC		5′-3,6-dihydrocytidylic acid	
S2C		2-thiocytidine	
FMC			
FRC		5-Formyl dC III CE	
FQC		Ac-5-Me-dC-CE	
FSC		5-Formyl-dC-CE	
FUC		5-Hydroxymethyl-dC II-CE	
FVC		5-I-dC-CE	5′-Dimethoxytrityl-N-benzoyl-5-iodo-2′-deoxyCytidine,3′-[(2-cyanoethyl)-

			(N,N-diisopropyl)]-phosphoramidite
FWC		5-Br-dC-CE	5′-Dimethoxytrityl-N-benzoyl-5-bromo-2′-deoxyCytidine,3′-[(2-cyanoethyl)-(N,N-diisopropyl)]-phosphoramidite
FYC		5-Carboxy-dC-CE	5′-Dimethoxytrityl-N-benzoyl-5-ethylcarboxy-2′-deoxyCytidine,3′-[(2-cyanoethyl)-(N,N-diisopropyl)]-phosphoramidite
FZC		5-Hydroxymethyl-dC-CE	5′-Dimethoxytrityl-N-benzoyl-5-cyanoethoxy-methyl-2′-deoxyCytidine,3′-[(2-cyanoethyl)-(N,N-diisopropyl)]-phosphoramidite
FTC		5-Me-dC-CE	5′-Dimethoxytrityl-N-benzoyl-5-methyl-2′-deoxyCytidine,3′

			-[(2-cyanoethyl)-(N,N-diisopropyl)]-phosphoramidite
Modified Guanine			
TIG		N2-trans-isosafrole-dG-adduct	
1NE		1,N2-ethenoguanine	
2NE		N2,3-ethenoguanine	
2MG		N2,N2-dimethylguanosine	
7MG		N2,7-dimethylguanosine	+1 charge
8OG		8-oxoguanine	
ADG		7-amido-7-deazaguanosine	
BGM		8-bromo-2′-deoxyguanosine	

		-5′-monophosphate	
BQG			
C7G		7-cyano-7-carbaguanine	
CEG		N2-carboxyethylguanine	
ISG		Iso-guanine	
M1G		1-methylguanine	
M2G		2-methylguanine	
M3G		3-methylguanine	
M6G		6-O-methylguanine	

M7G		7-methylguanine	+1 charge
NMG			
PQG		7-aminomethyl-7-carbaguanine	
TMG		N2,N2,7-trimethylguanosine	+1 charge
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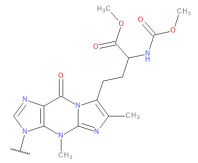
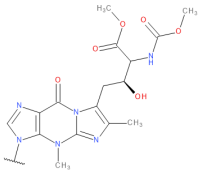
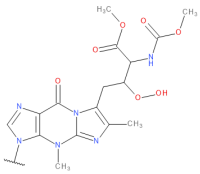
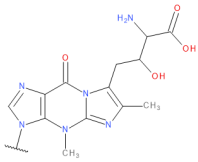
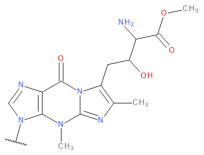
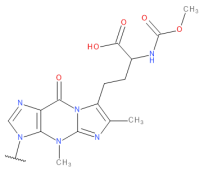
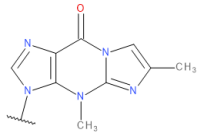
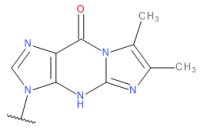
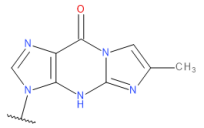
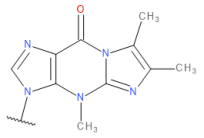
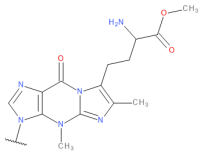
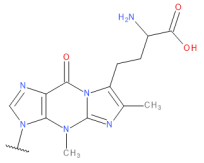
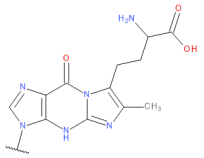
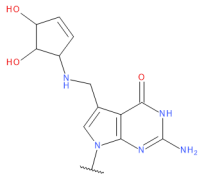
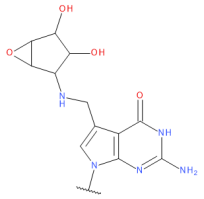
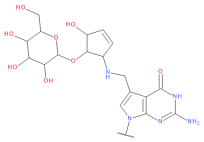
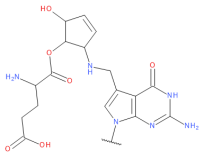
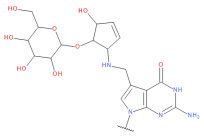
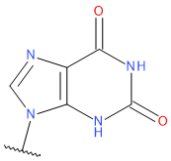
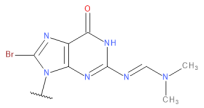
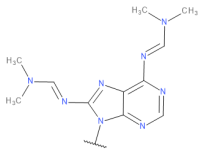
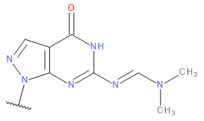
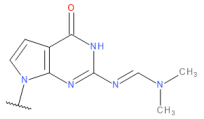
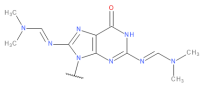
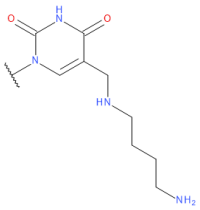
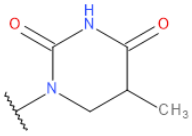
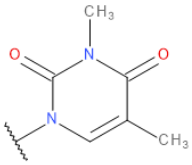
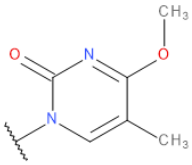
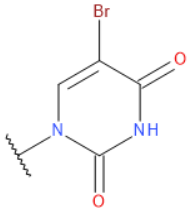
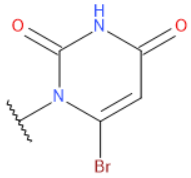
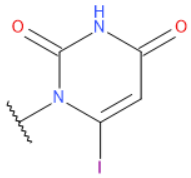
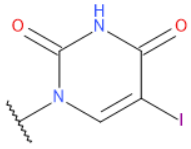
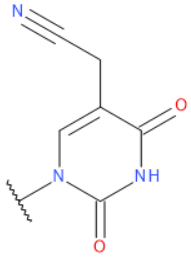
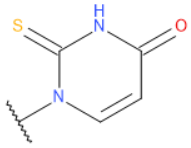
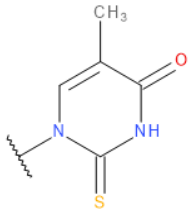
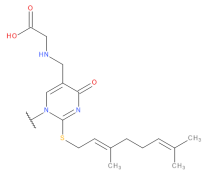
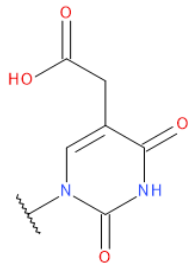
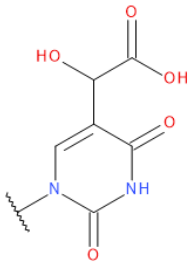
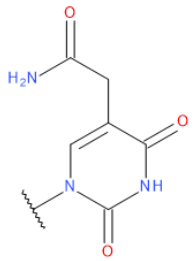
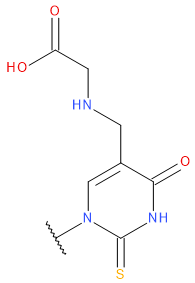
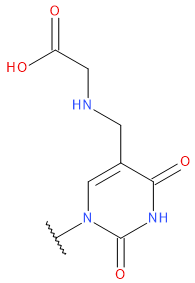
WYB		wybutosine	
HWB		hydroxywybutosine	
IPG		peroxywybutosine	
OHW		undermodified hydroxywybutosine	

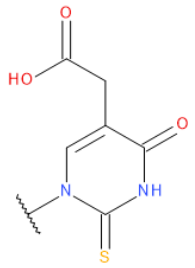
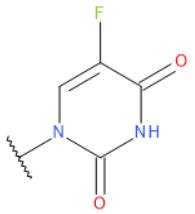
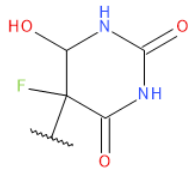
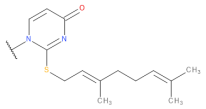
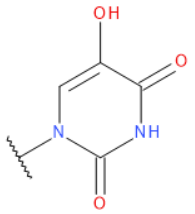
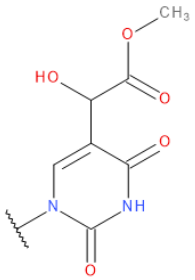
OMW		methylated undermodified hydroxywybutosine	
PYW		wybutosine[C15(S)]-5′-monophosphate	
IMG		wyosine	
IG2		isowyosine	
I4G		4-demethylwyosine	
MWG		methylwyosine	
YW5		7-aminocarboxypropylwyosine methyl ester	
YW7		7-aminocarboxypropylwyosine	
YW8		7-aminocarboxypropyl-demethylwyosine	

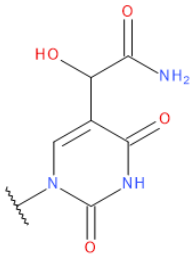
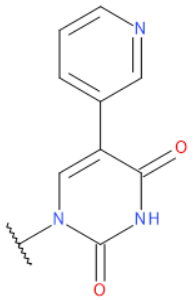
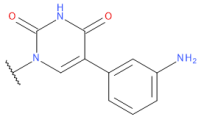
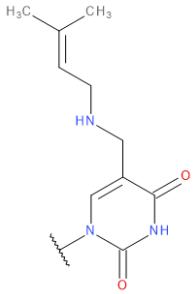
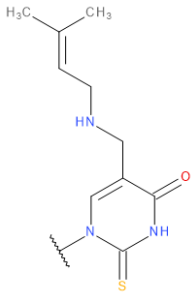
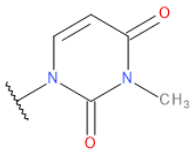
QUE		queuosine	
EOQ		epoxyqueuosine	
GAQ		galactosyl-queuosine	
GLQ		glutamyl-queuosine	
MQG		mannosyl-queuosine	
XAN		Xantine	
BRG		8-Br-dG-CE Phosphoramidite	
BSG		8-Amino-dA-CE Phosphoramidite	
BTH		7-Deaza-8-aza-dG-CE Phosphoramidite	
BTG		7-deaza-dG-CE	

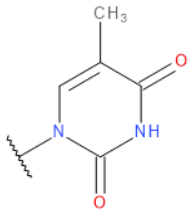
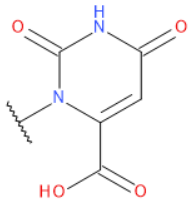
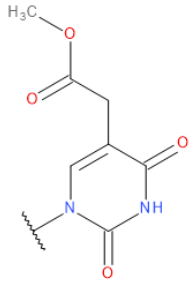
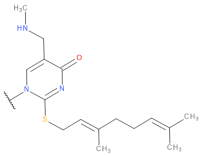
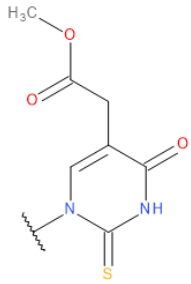
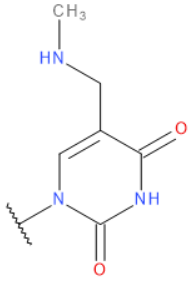
BUG		8-Amino-dG-CE	5′-Dimethoxytrityl-N2,N8-bis(dimethylaminomethylidine)-8-amino-2′-deoxyGuanosine, 3′-[(2-cyanoethyl)-(N,N-diisopropyl)]-phosphoramidite
Modified Thymine			
APT		alpha-putrescinylthymine	
DHT		5,6-dihydrothymine	
M3T		3-methylthymine	
O4T		O(4)-methylthymine	
Modified Uracil			
5BU		5-bromo-uridine	

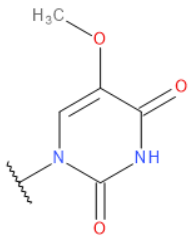
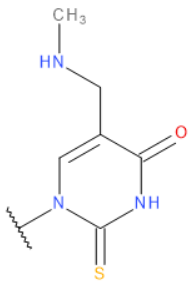
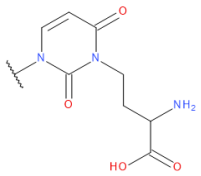
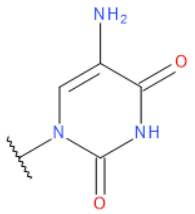
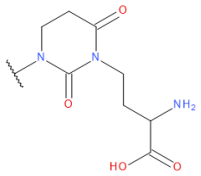
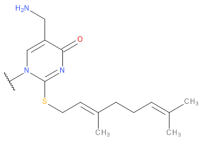
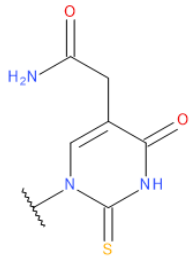
6BU		6-bromo-uridine	
I6U		6-iodouridine	
I5U		5-iodouridine	
5CU		5-cyanomethyluridine	
SSU		2-thiouridine	
5SU		5-methyl-2-thiouridine	
CGU		5-carboxymethylaminomethyl-2-geranylthiouridine	

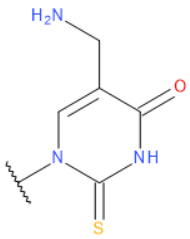
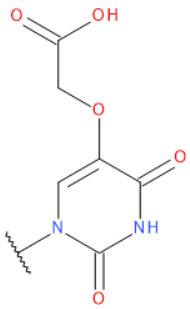
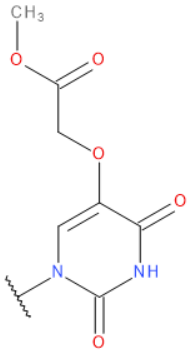
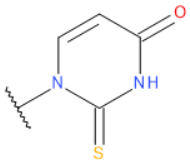
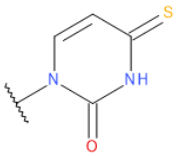
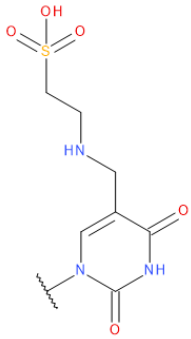
C5U		5-carboxymethyluridine	
CHU		5-carboxyhydroxymethyluridine	
CMU		5-carbamoylmethyluridine	
CNU		5-carboxymethylaminomethyl-2-thiouridine	
COU		5-carboxymethylaminomethyluridine	

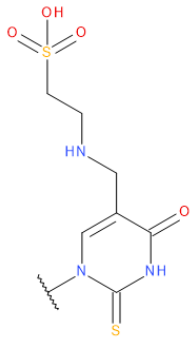
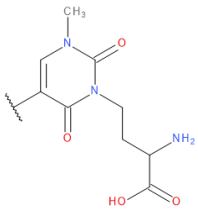
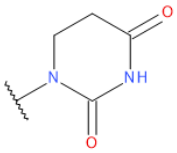
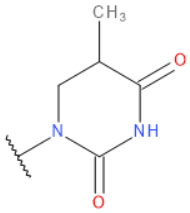
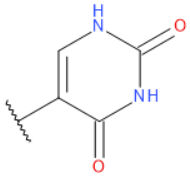
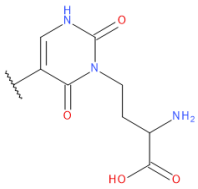
CSU		5-carboxymethyl-2-thiouridine	
FDU		5’-fluoro-uridine	
FHU		(5S,6R)-5-fluoro-6-hydroxy-pseudouridine-5′-monophosphate	
GSU		2-geranylthiouridine	
H5U		5-hydroxyuridine	
HMU		5-(carboxyhydroxymethyl)uridine methyl ester	

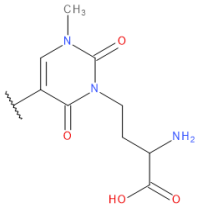
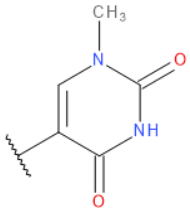
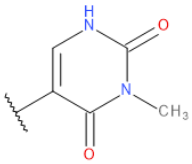
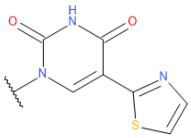
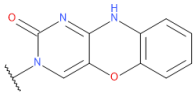
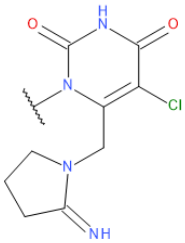
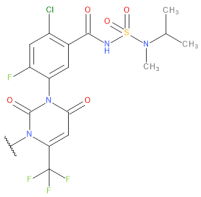
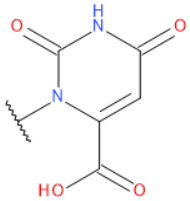
HOU		5-carbamoylhydroxymethyluridine	
BNC			
BOC			
INU		5-(isopentenylaminomethyl)uridine	
ISU		5-(isopentenylaminomethyl)-2-thiouridine	
M3U		3-methyluridine	

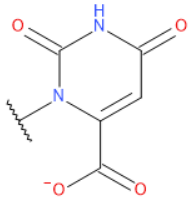
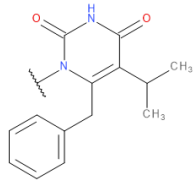
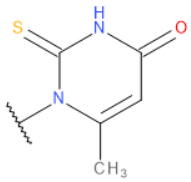
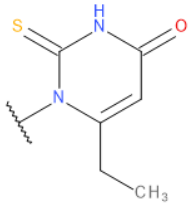
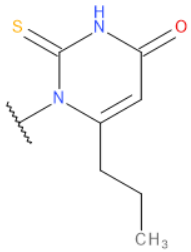
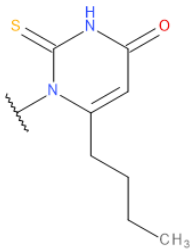
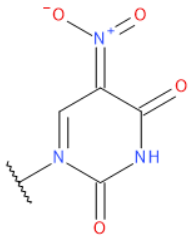
M5U		5-methyluridine	
M6U		uracil-6-ylacetic acid: DNAmod	
MCU		5-methoxycarbonyl methyluridine	
MGU		5-methylaminomethyl-2-geranylthiouridine	
MMU		5-methoxycarbonyl methyl-2-thiouridine	
MNU		5-methylaminomethyluridine	

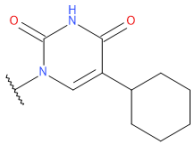
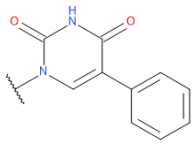
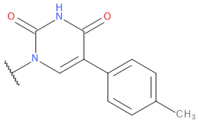
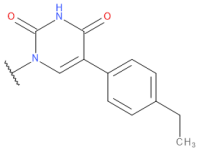
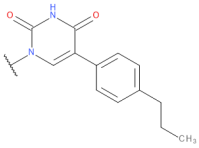
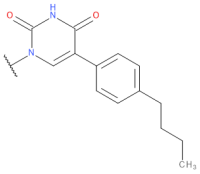
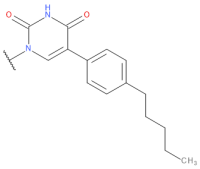
MOU		5-methoxyuridine	
MSU		5-methylaminomethyl-2-thiouridine	
N3U		3-(3-amino-3-carboxypropyl)uridine	
N5U		5-aminomethyluridine	
NCD		3-(3-amino-3-carboxypropyl)-5,6-dihydrouridine	
NGU		5-aminomethyl-2-geranylthiouridine	
NMU		5-carbamoylmethyl-2-thiouridine	

NSU		5-aminomethyl-2-thiouridine	
O5U		uridine 5-oxyacetic acid	
OMU		uridine 5-oxyacetic acid methyl ester	
S2U		2’-thiouridine	
S4U		4-thiouridine	
T5U		5-taurinomethyluridine	

TSU		5-taurinomethyl-2-thiouridine	
B8N		(2~{R})-2-azanyl-4-[5-[(2~{S},3~{R},4~{S},5~{R})-3,4-bis(oxidanyl)-5-(phosphonooxymethyl)oxolan-2-yl]-3-methyl-2,6-bis(oxidanylidene)pyrimidin-1-yl]butanoic acid	
BDD		Dihydrouridine	
M5D		5-methyldihydrouridine	
PUU		Pseudouridine	
C3Y		3-(3-amino-3-carboxypropyl)pseudouridine	

MCY		1-methyl-3-(3-amino-3-carboxypropyl)pseudouridine	
M1Y		1-methylpseudouridine	
M3Y		3-methylpseudouridine	
GXT			
GYT			
TIP		Tipiracil	
SFF		Saflufenacil	
OAU		Orotic acid	

OBU		Orotate	
EMU		Emivirine	
PVU		6-methyl-2-thiouracil	
PSU		6-ethyl-2-thiouracil	
PTU		6-propyl-2-thiouracil	
PWU		6-butyl-2-thiouracil	
5NU		5-nitrouracil	

RAU		5-cyclohexane-uracil	
RBU		5-phenyl-uracil	
RCU		p-methylphenyl-5-uracil	
RDU		p-ethylphenyl-5-uracil	
REU		p-propyl phenyl-5 uracil	
RFU		C=4	
RGU		C=5	
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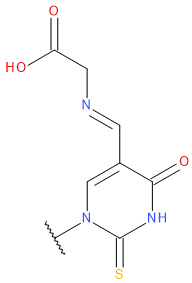
1RN		(E)-N-{[4-oxo-1-(5-O-phosphono-beta-D-arabinofuranosyl)-2-thioxo-1,2,3,4-tetrahydropyrimidin-5-	

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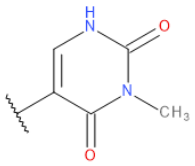
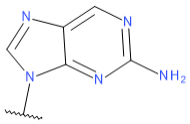
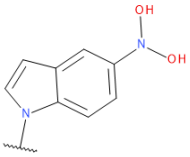
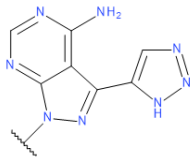
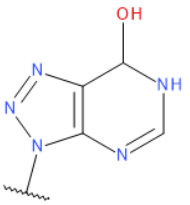
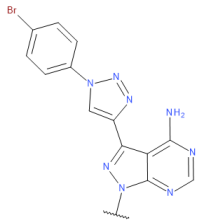
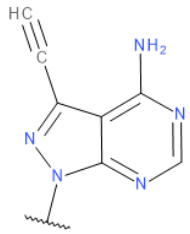
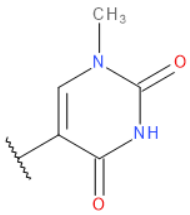
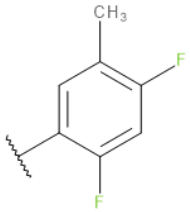
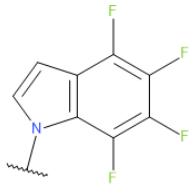
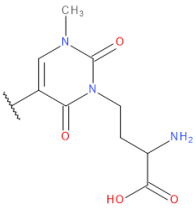
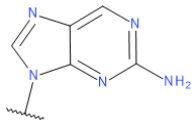
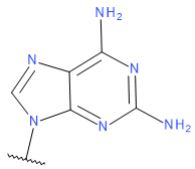
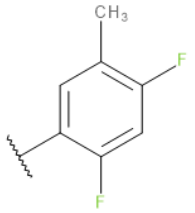
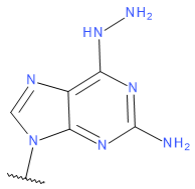
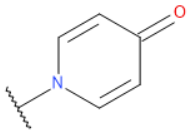
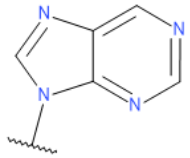
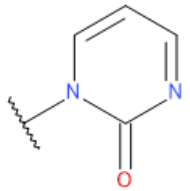
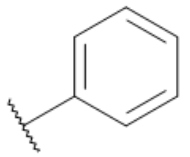
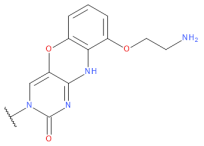
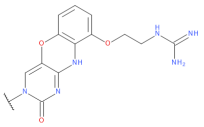
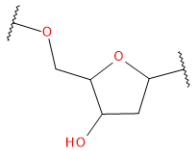
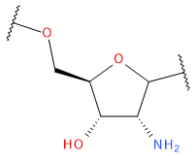
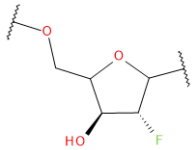
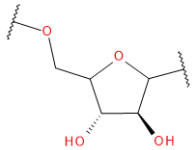
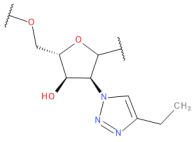
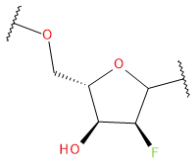
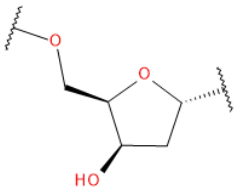
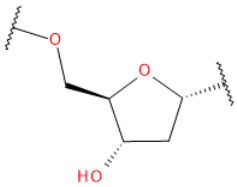
3TD		(1S)-1,4-anhydro-1-(3-methyl-2,4-dioxo-1,2,3,4-tetrahydropyrimidin-5-yl)-5-O-phosphono-D-ribitol	
2PR		2-amino-9-[2-deoxyribofuranosyl]-9H-purine-5′-monophosphate	
5NI		5’-nitroindole	
7AT		1-(5-O-phosphono-beta-D-ribofuranosyl)-3-(1H-1,2,3-triazol-5-yl)-1H-pyrazolo[3,4-d]pyrimidin-4-amine	
8AZ		8-aza-nebularine-5′-monophosphate	

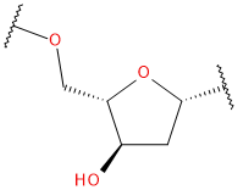
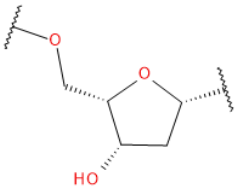
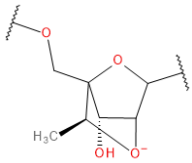
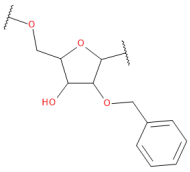
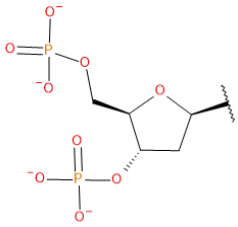
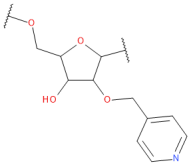
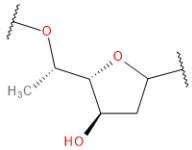
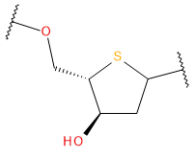
A7C		3-[1-(4-bromophenyl)-1H-1,2,3-triazol-4-yl]-1-[5-O-(trihydroxy-lambda~5~-phosphanyl)-beta-D-ribofuranosyl]-1H-pyrazolo[3,4-d]pyrimidin-4-amine	
A7E		3-ethynyl-1-(5-O-phosphono-beta-D-ribofuranosyl)-1H-pyrazolo[3,4-d]pyrimidin-4-amine	
B8H		[(2~{R},3~{S},4~{R},5~{S})-5-[1-methyl-2,4-bis(oxidanylidene)pyrimidin-5-yl]-3,4-bis(oxidanyl)oxolan-2-yl]methyl dihydrogen phosphate	
BRF		2’,4’-difluorotoluyl ribonucleoside	

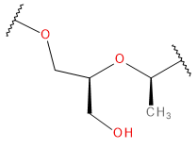
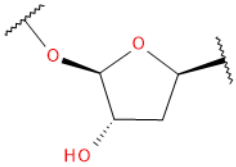
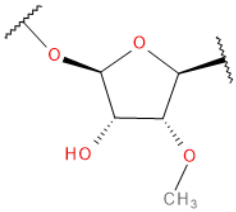
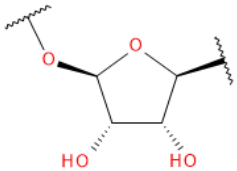
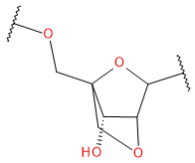
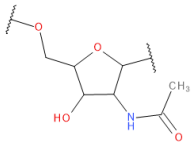
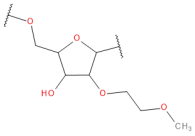
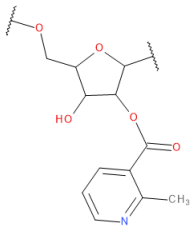
F4F			
C4J		(5S)-5-{3-[(3S)-3-amino-3-carboxypropyl]-1-methyl-2,4-dioxo-1,2,3,4-tetrahydropyrimidin-5-yl}-2,5-anhydro-1-O-phosphono-L-arabinitol	
MTU		9-beta-D-ribofuranosyl-9H-purin-2-amine	
N6G		((2R,3S,4R,5S)-5-(2,6-diamino-9H-purin-9-YL)-3,4-dihydroxy-tetrahydrofuran-2-YL)methyl dihydrogen phosphate	
NF2		(1S)-1,4-anhydro-1-(2,4-difluoro-5-methylphenyl)-5-O-phosphono-D-ribitol	

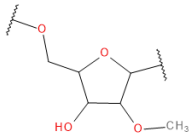
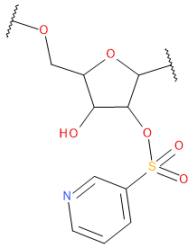
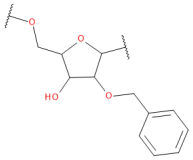
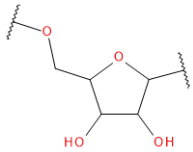
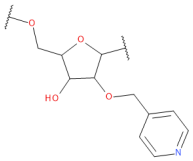
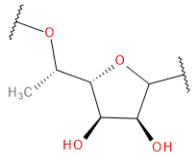
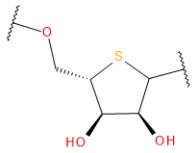
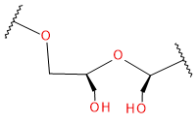
O2Z		[(2~{R},3~{S},4~{R},5~{R})-5-(2-azanyl-6-diazanyl-purin-9-yl)-3,4-bis(oxidanyl)oxolan-2-yl]methoxyphosphinic acid	
ONE		1-(beta-D-ribofuranosyl)-pyridin-4-one-5′-phosphate	
P5P		purine riboside-5′-monophosphate	
PYO		1-(beta-D-ribofuranosyl)-pyrimidin-2-one-5′-phosphate	
PYY		D-ribofuranosyl-benzene-5′-monophosphate	
GCL			
GC2			

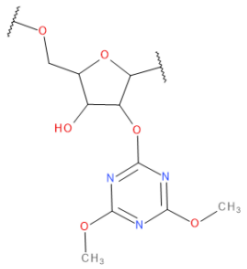
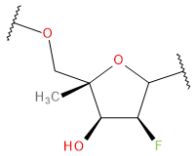
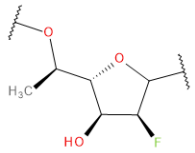
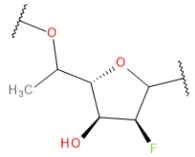
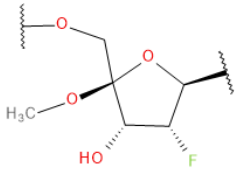
Sugar/linkers

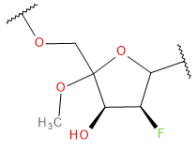
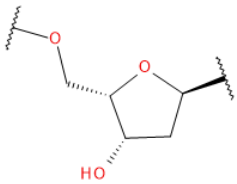
FRAGMENT ID	Structure	Name	Notes
DC2		Standard deoxyribose	Atom types from OL15
2AD		2’amino-2’deoxy	
A5L		(2-deoxy-2-fluoro-5-O-phosphono-beta-D-arabinofuranosyl)	2’FANA, built C2’endo
A5O		arabinose	*note the 2’OH group points “up” instead of “down”
A9Z		2′-(4-ethyl-1H-1,2,3-triazol-1-yl)	
AF2		2′-deoxy-2′-fluoro	Built C3’endo
aDX			
aXX			

bLD			
bYX			
CET		“gapmer”	https://pubs.acs.org/doi/abs/10.1021/jo502320y
DBO		2′-O-benzyl	built-c2’endo
DDD			
DMP		2′-O-methyl-4-pyridine	built-c2’endo
DMX			built-c2’endo
DS4		4′-thio-4′-deoxy-cytosine-5′-monophosphate, S4C	

DUD		Unlocked nucleic acid	Built C2'endo without OH
GDS			
GMS			
GRS			
LNA		Locked nucleic acid	Built C2'endo (but C3'endo also optimizes to C2'endo)
M5M		2′-(N-acetamide)-cytidine-5′-monophosphate	
MOE		2′-O-methoxyethyl	built-C3'endo only
NAI			Built C2'endo

OME		2′-O-methyl	
P3S			Sulfonyl ester on O2′, Built C2’endo
RBO		2′-O-benzyl	Built C3’endo
RC3		Standard ribose sugar	C3’endo, atom types from OL3
RMP		2′-O-methyl-4-pyridine	Built C3’endo
RMX		2-amino-9-(6-deoxy-5-O-phosphono-beta-D-allofuranosyl)-3,9-dihydro-6H-purin-6-one	Built C3’endo
RS4		S4C, 4′-thio-4′-deoxy-cytosine-5′-monophosphate	
RUR		Unlocked nucleic acid	Built c3’endo with 2’OH

TAE			Triazole aryl ether on O2′, Built with C2′endo
U4M		1-{2,5-dideoxy-2-fluoro-4-[(phosphonooxy)methyl]-alpha-L-lyxofuranosyl}pyrimidine-2,4(1H,3H)-dione	Methyl group on C4′, fluorine on C2′
U5M		1-(2,6-dideoxy-2-fluoro-5-O-phosphono-alpha-L-talofuranosyl)pyrimidine-2,4(1H,3H)-dione	Methyl group on C5′, fluorine on C2′
U5R		1-(2,6-dideoxy-2-fluoro-5-O-phosphono-beta-D-allofuranosyl)pyrimidine-2,4(1H,3H)-dione	D-enantiomer version of U5M
UFB			

UMO		2′-F-4′-OMe uridine 5′-monophosphate	
YLX			

ModXNA

A tool to create new nucleotides for AMBER